Hoff Bayesian Statistics

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These are (fully reproducible!) R Markdown lecture notes for Hoff (2009), completed as part of a 1-semester independent study course. Only Chapters 1-8 are complete right now.

Each note includes summaries of chapter sections, with math and explanations modified for clarity and the occasional link to external resources. Many figures from the book are reproduced in a ggplot/tidyverse style, and some exercises at the end of each chapter are tackled (correctness not guaranteed).

If you find an error or would like to improve the notes, please submit a PR or open an issue!

1 Chapters

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2 Final Project

As a small final project, R code for the basic binary relation version of the Infinite Relational Model was implemented, as described in Kemp et al. (2006).

- Infinite Relational Model[11]

Hoff, Peter D. 2009. *A First Course in Bayesian Statistical Methods*. Springer Texts in Statistics. Springer. https://doi.org/10.1007/978-0-387-92407-6.

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Kemp, Charles, Joshua B. Tenenbaum, Thomas L. Griffiths, Takeshi Yamada, and Naonori Ueda. 2006. "Learning Systems of Concepts with an Infinite Relational Model." *Proceedings of the National Conference on Artificial Intelligence (AAAI)* 21: 381–88. http://web.mit.edu/cocosci/Papers/Kemp-etal-AAAI06.pdf.